

Backfill | Working candidate v1

Frozen 4 October 2026. Post-result exploratory research. Retain the current candidate; neither fixed alternative passed the replacement gate. This does not replace or validate the registered primary.

Rule and execution

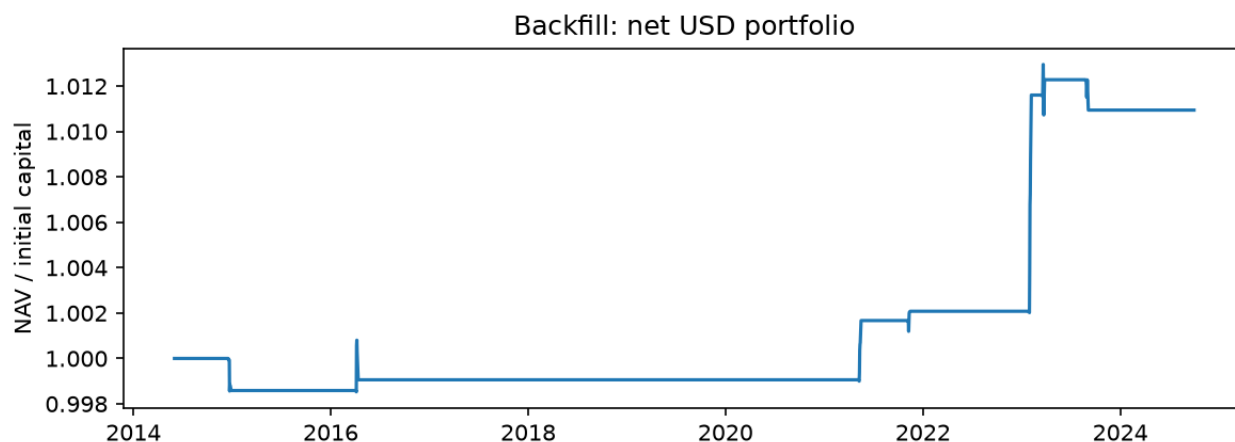
US listed owner; every FDA-listed presentation on the selected archived formulation page available, unallocated and nonempty. After both shortage and supplier evidence are available, identify the next US close. **Enter 20 additional sessions later; sell five sessions after entry.** No SMA filter. Supplier status is not reverified during the wait.

Equal 5% capital per event, split before exclusions; unused shares stay cash. Hedge each lot with SPY using 250 prior paired returns. Stock 10bp/SPY 2bp each side; hedge carry 50bp/year. Cash earns zero. Costs and carry doubled in stress.

Net in-sample evidence | June 2014-September 2024

Metric	Historical reference	Webull-only
Lots / events	8 / 7	4 / 3
Sharpe	0.535	0.609
Sharpe, doubled costs	0.495	0.591
Annualized return	+0.105%	+0.100%
Annualized volatility	0.198%	0.165%
Max drawdown	-0.22%	-0.22%
Annual turnover	12.87%	5.23%
Mean net lot return	+2.84%	+5.40%

The reference contains four Yahoo-priced TEVA trades. Webull-only keeps those allocations in cash. Annual figures include inactive sessions and should not be read as typical returns on invested capital.



Evidence and actual trades

The frozen strict ledger has 18 all-market candidates across 15 events. The US subset has 11 candidates; three MYL candidates are excluded because predecessor prices are unavailable. Eight reference lots remain across seven events and five owners. No successor history substitutes for a delisted predecessor.

Actual reference fills and net lot returns

Owner	Event	Buy close	Sell close	Net lot
TEVA	93	2014-12-19	2014-12-29	-2.83%
PFE	131	2016-04-05	2016-04-12	+0.95%
TEVA	338	2021-05-10	2021-05-17	+5.23%
BAX	352	2021-11-08	2021-11-15	+0.92%
ICUI	352	2021-11-08	2021-11-15	+0.72%
AMRX	408	2023-01-30	2023-02-06	+19.01%
TEVA	413	2023-03-20	2023-03-27	+1.34%
TEVA	430	2023-08-28	2023-09-05	-2.65%

Lot returns reconcile stock and hedge cashflows, both-side transaction costs and hedge carry. They measure return on allocated lot capital; the portfolio metrics also account for cash and overlapping lots.

What the evidence does and does not show

Supplier status comes from one point-in-time FDA formulation page within the 30-day rule. All parent-owned presentation rows on that page qualify. This does not establish that the manufacturer's whole catalogue is available, that it has spare capacity, or that another owner's strength/route is commercially substitutable.

A disrupted counterpart requires a separately listed owner on the exact same formulation capture. Different archive timestamps and the same owner are rejected. Page-absent generic-maker controls use the same event dates; absence does not prove they do not manufacture the product.

Some shortage observation dates remain uncertain. Entry uses evidence availability rather than a guessed earlier onset. Archived status can be stale at the delayed entry; the frozen rule does not manufacture an update.

Bounded improvement check

Two alternatives were fixed before this run: immediate entry when the preceding close is below its trailing 60-session average; and the same rule requiring a documented disrupted listed counterpart. Both hold five sessions and keep the same frozen event allocations.

Policy / entry rule	Lots	Sharpe	x2 costs	Mean net lot
Reference / Wait 20	8	0.535	0.495	+2.84%
Reference / SMA60 immediate	3	0.087	0.061	+0.86%
Reference / SMA60 + disruption	1	0.411	0.399	+9.61%
Webull / Wait 20	4	0.609	0.591	+5.40%
Webull / SMA60 immediate	2	0.390	0.367	+4.81%
Webull / SMA60 + disruption	1	0.411	0.399	+9.61%

Neither alternative passes. The disruption condition leaves only AMRX. Replacement required higher Sharpe at both costs on the same vendor policy, positive winner-minus-control mean, no worse drawdown, ten unique exposures across three owners, and positive retained-lot means after every company omission. The current candidate also lacks ten exposures; keeping it is a working decision, not certification.

Already-tested nearby timing | reference Sharpe

Extra entry sessions	Hold 1	Hold 5	Hold 20
0	-0.232	0.303	0.024
5	-0.073	-0.100	0.094
20	0.401	0.535	0.105

AMRX contributes 87.0% of net dollar profit. Without it, mean net lot return is +0.52% and Sharpe 0.104. Excluding entry-year 2023 leaves Sharpe 0.205. Omitted allocations stay cash. The relation is weak without its main contributor.

Inference, factors and drawdowns

Uncertainty and research history

Calendar-time winner HAC(60) $t=1.08$, $p=0.279$. Winner-minus-control $p=0.207$. Mean-lot descriptive bootstrap interval: -0.86% to $+8.23\%$, resampling seven information-date clusters together. These selected-sample intervals are not confirmatory or adjusted for choosing the timing.

Preserved strict timing research contains 42 strategy cells, 168 winner/control evaluations and 38 omission diagnostics. Its log records 111 earlier inference probes. A separate sample tested nine entry/hold rules. This package adds two entry/filter alternatives, two vendor policies and two costs: 24 winner/control evaluations. Identical reruns corrected a serialization failure and verified the final audit; all attempts remain recorded. Known lower-bound multiplicity-adjusted winner p-values are all 1.

Mean descriptive CAR over 20 sessions strictly before all evidence was available is -3.04% . Beta is fitted using only pre-information prices. This is not flat and does not establish that the pre-drift prediction passed. The existing pre-entry diagnostic is retained separately because a delayed entry's lookback can include post-announcement days.

US monthly factor check | 124 months, HAC(3)

Factor / intercept	Coefficient	t-stat
Market	-0.00122	-0.42
Value	0.00236	0.65
Momentum	-0.00208	-0.60
Monthly alpha	-0.00115	-4.46

Official Ken French current factor vintage; retrospective exposure check, never a trading input. Regression subtracts its risk-free rate. Negative alpha partly reflects zero-return idle cash versus that benchmark; positive Sharpe above zero is not evidence of positive cash-benchmark alpha.

Only 35 of 2601 sessions have positions. Worst month -0.14% ; max drawdown -0.22% . No 10% de-risk trigger. Active-year portfolio returns: 2014 -0.141% , 2016 $+0.047\%$, 2021 $+0.302\%$, 2023 $+0.885\%$. Sparse exposure prevents persuasive regime or tail-risk claims.

The previously evaluated US primary remains negative: 19 lots / 17 events, Sharpe -0.594 , annual return -0.56% , max drawdown -6.66% , HAC $p=0.156$. Refinement followed that result; it does not confirm the original hypothesis.

Risk, capacity and reproducibility

Caps: 8% name, 30% listing market, 50% healthcare, 150% gross, 25% absolute net. Independent lots retain their expiry. A 10% drawdown halves exposure at the next session; restore after 20 sessions. No early shortage-resolution exit. All stocks in this candidate are US listings, valued in USD.

Observed max marked name 5.78%, gross 10.76%, absolute net 2.77%. Price drift can move marked exposure above an entry-sized cap; none here breaches its cap.

Liquidity and modeled execution

ADV is average dollar volume over 60 preceding sessions. For each date and symbol, sum absolute lot orders across both legs. Capacity AUM = participation limit x ADV / order fraction of NAV. The minimum across entries and exits is **\$461,086 at 1% ADV**, or \$2,305,430 at 5%. AMRX exit is limiting. \$1m is an accounting normalization; its AMRX exit is about 2.19% ADV.

Corwin-Schultz two-day ranges give AMRX full-spread estimates near 89bp. The frozen 10bp-per-side stock model, even doubled, is below this indicative spread. Treat reported performance as a fixed-cost benchmark pending calibration. These volume bounds are not profitable capacity estimates. Webull adjustment/dividend and volume conventions are not independently certified; zero estimated spread does not establish free execution.

Audit and reproduction

Frozen ledgers, all-presentation audits, price/factor manifests, source/code hashes and dedicated lifecycle logs are included. Save requests, rejected entries, eligibility, trades, lot cashflows, daily equity, metrics, factors, pre-information drift and capacity. The reference reproduces exactly and every completed lot ledger reconciles with NAV. All 27 synthetic/guardrail tests pass. No new stock prices or holdout evaluation were used.

Run `.venv/bin/python -m research.final_candidate.run` with the local frozen cache. Missing or altered files stop execution. Raw vendor prices remain gitignored; reproducing numerical results requires that licensed cache. Make a new disclosed version to edit rules; keep v1 and all prior results.

What remains unconfirmed

The 2024-2026 period has already been inspected; this candidate has no new blind OOS result. No final command or lock reset occurred. A genuinely unseen cohort and calibrated costs are needed before calling this a validated edge. This is a five-page research note, not a complete competition submission with a certified unseen test.

Sources: frozen FDA evidence and quote manifest; official Ken French Data Library (mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html). Detailed CSVs and hashes accompany the report in `results/research/final_candidate_v1_20261004`.